Polynomial equations

Lecture and worked examples

Objectives

Solve in $\mathbb{C}$ a quadratic equation with real coefficients; factorise a polynomial by $z - a$ when a is a root; solve a cubic equation when one root is known; know that a polynomial of degree n has at most n roots.

Introductory activity

Consider $P(z) = z^3 - 2z^2 + 4z - 8$.

Introductory activity (2)

1. Plot on a calculator the curve of $x \mapsto x^3 - 2x^2 + 4x - 8$ on $[-3, 4]$. How many real solutions does the equation $P(x) = 0$ seem to have? Locate one.
2. Check that $P(2) = 0$.
3. Find real numbers a, b, c such that, for every z, $P(z) = (z-2)\left(az^2 + bz + c\right)$. Expand the right-hand side and identify the coefficients.
4. Deduce all the complex solutions of $P(z) = 0$.
5. Compute the sum and the product of the three solutions. Compare with the coefficients of P.

Introductory activity (3)

6. How many complex solutions can a polynomial of degree 3 have at most? State a conjecture for degree n.

Introductory activity (4)

Observing is not proving: the properties conjectured here are established in the course.

Quadratics with real coefficients

Let a, b, c be three real numbers with $a \neq 0$, and $\Delta = b^2 - 4ac$. The equation $az^2 + bz + c = 0$ has:

1. if $\Delta > 0$, two real solutions $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$;
2. if $\Delta = 0$, one real solution $-\dfrac{b}{2a}$;
3. if $\Delta < 0$, two conjugate complex solutions $z_1 = \dfrac{-b - i\sqrt{-\Delta}}{2a}$ and $z_2 = \dfrac{-b + i\sqrt{-\Delta}}{2a}$.

Quadratics with real coefficients (2)

In every case, $az^2 + bz + c = a(z - z_1)(z - z_2)$, $z_1 + z_2 = -\dfrac{b}{a}$ and $z_1 z_2 = \dfrac{c}{a}$ (with $z_1 = z_2$ if $\Delta = 0$).

Quadratics with real coefficients

The completed square form $az^2 + bz + c = a\left[\left(z + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}\right]$ still holds in $\mathbb{C}$. If $\Delta < 0$, write $\frac{\Delta}{4a^2} = \left(\frac{i\sqrt{-\Delta}}{2a}\right)^2$, then factorise the difference of two squares. A product of complex numbers is zero if and only if one of the factors is zero (since every nonzero complex number is invertible). Sum and product are obtained by expanding $a(z - z_1)(z - z_2)$ and identifying.

Quadratics with real coefficients

$\Delta = 4 - 20 = -16 = (4i)^2$, so $z = \dfrac{2 \pm 4i}{2}$: the solutions are $1 - 2i$ and $1 + 2i$.

Factorisation by $z - a$

A *polynomial function* of degree n is a function $P : z \mapsto a_n z^n + \cdots + a_1 z + a_0$ defined on $\mathbb{C}$, where $a_0, \ldots, a_n$ are complex numbers and $a_n \neq 0$. A complex number α such that $P(\alpha) = 0$ is a *root* of P.

Factorisation by $z - a$

If a polynomial function is zero for every complex number z, then all its coefficients are zero. Consequently, two equal polynomial functions have the same coefficients: we may *identify* the coefficients.

Factorisation by $z - a$

For all complex numbers z, a and every integer $n \geq 1$:

$$z^n - a^n = (z-a)\left(z^{n-1} + az^{n-2} + \cdots + a^{n-2}z + a^{n-1}\right) = (z-a)\sum_{k=0}^{n-1} a^k z^{n-1-k}.$$

Factorisation by $z - a$

Expand: $(z-a)\sum_{k=0}^{n-1} a^k z^{n-1-k} = \sum_{k=0}^{n-1} a^k z^{n-k} - \sum_{k=0}^{n-1} a^{k+1} z^{n-1-k}$. Setting $j = k+1$ in the second sum, it becomes $\sum_{j=1}^{n} a^j z^{n-j}$. The terms with index 1 to $n-1$ cancel; what remains is the term $k = 0$ of the first sum, z^n, and the term $j = n$ of the second, a^n. We obtain $z^n - a^n$.

Factorisation by $z - a$

Let P be a polynomial function of degree $n \geq 1$ and α a root of P. There is a polynomial function Q of degree $n - 1$ such that, for every complex number z, $P(z) = (z - \alpha)Q(z)$.

Factorisation by $z - a$

Write $P(z) = \sum_{k=0}^{n} a_k z^k$. Since $P(\alpha) = 0$,

$$P(z) = P(z) - P(\alpha) = \sum_{k=1}^{n} a_k \left(z^k - \alpha^k\right).$$

By the previous theorem, each $z^k - \alpha^k$ can be written $(z-\alpha)Q_k(z)$ with Q_k a polynomial of degree $k-1$ with leading coefficient 1. So $P(z) = (z-\alpha)Q(z)$ with $Q = \sum_{k=1}^{n} a_k Q_k$, of degree $n-1$ with leading coefficient $a_n \neq 0$.

Factorisation by $z - a$

A polynomial function of degree $n \geq 1$ has at most n roots.

Factorisation by $z - a$

By induction on n. A polynomial function of degree 1, $a_1 z + a_0$ with $a_1 \neq 0$, has exactly one root, $-\dfrac{a_0}{a_1}$. Assume the property holds in degree n, and let P have degree $n+1$. If P has no root, we are done. Otherwise, let α be a root: $P(z) = (z - \alpha)Q(z)$ with Q of degree n. If $P(\beta) = 0$, then $\beta = \alpha$ or $Q(\beta) = 0$. Since Q has at most n roots, P has at most $n+1$.

Factorisation by $z - a$

If P has real coefficients and α is a root of P, then $\overline{\alpha}$ is also a root of P.

Factorisation by $z - a$

Since the coefficients are real, $P(\overline{\alpha}) = \sum a_k \overline{\alpha}^k = \overline{\sum a_k \alpha^k} = \overline{P(\alpha)} = 0$.

Factorisation by $z - a$

Factorise by $z - \alpha$ by identifying the coefficients of $Q(z) = az^2 + bz + c$, then solve $Q(z) = 0$. Example: $z^3 + 8 = 0$ has the root -2 and $z^3 + 8 = (z + 2)\left(z^2 - 2z + 4\right)$. The quadratic has discriminant $-12 = \left(2i\sqrt{3}\right)^2$, hence the solutions -2, $1 - i\sqrt{3}$, $1 + i\sqrt{3}$.

Factorisation by $z - a$

The discriminant of a quadratic with real coefficients no longer tells whether the equation has solutions in $\mathbb{C}$: it always has. It tells whether they are real or conjugate complex numbers.